

Effects of α -ketoglutarate on growth performance, antioxidant capacity and ammonia metabolization against chronic carbonate alkalinity stress in Songpu mirror carp (*Cyprinus carpio* Songpu)

Ze Fan¹  | Feng Peng¹ | Jinnan Li¹ | Di Wu¹ | Yuanyuan Zhang¹ | Chang'an Wang¹ | Liansheng Wang¹ | Qiyu Xu^{1,2} 

¹Key Laboratory of Aquatic Animal Diseases and Immune Technology of Heilongjiang Province, Heilongjiang River Fisheries Research Institute, Chinese Academy of Fishery Sciences, Harbin, China

²School of Life Science, Huzhou University, Huzhou, China

Correspondence

Liansheng Wang and Qiyu Xu, Key Laboratory of Aquatic Animal Diseases and Immune Technology of Heilongjiang Province, Heilongjiang River Fisheries Research Institute, Chinese Academy of Fishery Sciences, Harbin 150070, China. Emails: wangliansheng@hrfri.ac.cn (LW); xuqiyu@sina.com (QX)

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Abstract

A 3 × 2 factorial experiment was conducted to investigate the interaction between carbonate alkalinity and dietary α -ketoglutarate (AKG) levels on the growth performance, antioxidant capacity and ammonia metabolization of Songpu mirror carp (*Cyprinus carpio* Songpu). Each diet (0%, 1% AKG) was randomly allotted to 0 mmol/L, 15 mmol/L, 30 mmol/L carbonate alkalinity groups with three replicate aquaria. The weight gain rate (WGR) significantly increased in the 1% AKG group and significantly decreased with increasing carbonate alkalinity ($p < .05$). Crude ash was significantly affected by the interaction between carbonate alkalinity and dietary AKG levels and significantly increased in the 1% AKG group ($p < .05$). A significant interaction between carbonate alkalinity and dietary AKG levels on superoxide dismutase (SOD) activities in hepatopancreas was observed ($p < .05$). The increasing carbonate alkalinities significantly decreased SOD activities in the hepatopancreas and foregut, catalase (CAT) activities in the hepatopancreas, foregut, midgut and hindgut and glutathione (GSH) contents in the hepatopancreas and foregut ($p < .05$). Despite AKG addition, the blood ammonia content still increased with increasing carbonate alkalinities. The gene expressions of AQP3, Rhag, Rhcg2 and Na⁺/K⁺-ATPase (NKA) in the gills and glutamine synthase (GS) in the brain were significantly upregulated with increasing carbonate alkalinities ($p < .05$). In the 1% AKG group, GS gene expression significantly upregulated in the brains, whereas AQP3, Rhag, Rhcg2 and NKA gene expressions significantly downregulated in the gills ($p < .05$). In summary, the 1% AKG addition can enhance the gene expression of ammonia metabolization and improve the antioxidant capacity of Songpu mirror carp with chronic carbonate alkalinity stress.

KEYWORDS

α -ketoglutarate, ammonia metabolization, antioxidant capacity, carbonate alkalinity, growth performance, Songpu mirror carp

1 | INTRODUCTION

As an important index of the water environment, carbonate alkalinity plays an essential role in the growth process. Alarming, it is hard for fish to draw the accurate range of carbonate alkalinity tolerance (Yao, Lai, Zhou, & Rizalita, 2010). In the 1980s, related researchers pointed out that low levels of carbonate alkalinity in the water environment would restrict fish productivity; therefore, the carbonate alkalinity in the aquarium was preferably in the range of 1–3 mmol/L (Zhanjiang fisheries college, 1980). During the same period, other researchers emphasized that the influence of pH on carbonate alkalinity toxicity to fish should not be neglected, and there is a positive correlation between carbonate alkalinity toxicity and pH in the water environment (Lei, Dong, & Chen, 1985). According to the early literature, the half lethal concentration of carbonate alkalinity in silver carp, bighead carp and grass carp was 10 mmol/L when pH in the aquarium reached 9.6. Generally, the pH of 9.6 is the high limit value in the aquarium (Lei et al., 1985). However, considering the buffer performance enhancement and relatively stable pH in a high-concentration carbonate alkalinity environment, pH will not easily reach above 9.6. Hence, related researcher suggested that total carbonate alkalinity of 10 mmol/L was recommended as a risk indicator in the aquarium. As the saline-alkaline fisheries further develops in recent years, more and more researches about fish have been undergone on the range and mechanism of carbonate alkalinity tolerance. Zhou obtained results from the study on the crucian carp *Carassius auratus* Linnaeus in Dali Lake in which the values of LC50 of carbonate alkalinity were 71.93 mmol/L at 24 hr, 67.80 mmol/L at 48 hr, 65.80 mmol/L at 72 hr and 63.42 mmol/L at 96 hr with the safe alkalinity of 18.07 mmol/L under the conditions of salinity 1.9–4.3 and pH 8.5 ± 0.5 (Zhou et al., 2013). When *Leuciscus waleckii* juveniles were acclimated to alkalinities of 18 mmol/L, 30 mmol/L and 50 mmol/L for 22 d and 60 d, total plasma-free amino acid contents increased with alkalinity and exposure time, suggesting that amino acids play an important role as an energy source during acclimation of *L. waleckii* to high alkalinity conditions (Chang, He, Sun, Liang, & Sun, 2016).

Excessive carbonate alkalinity in the water environment will dramatically reduce digestive enzyme activity, thereby reducing the feed ration, feed efficiency and growth rate of fish. The mechanism of carbonate alkalinity stress on fish is complex and involves many aspects (Roy, Davis, Saoud, & Henry, 2007; Yao et al., 2010). It is noteworthy, however, that one of the most important aspects is related to the obstruction of ammonia (NH_3) excretion in fish. In a high-concentration carbonate alkalinity environment, the pH value inside the body tissues rises because the acid-base equilibrium of the body is destroyed. Meanwhile, the amount of available H in the body substantially decreases as the pH value inside the body tissues rises, ultimately destroying the equilibrium system of NH_3 and NH_4^+ and leading to the obstruction of ammonia excretion from the body (Wood, Perry, Wright, Bergman, & Randall, 1989). As ammonia accumulates, the concentration of ammonia in

vivo gradually increases. Accumulation of ammonia in the blood eventually leads to serious reductions in respiratory function and blood oxygen-carrying capacity and damages the liver and kidney metabolic systems, nervous system, immune system and antioxidant system, resulting in ammonia poisoning (Ching, Chew, Wong, Yuen, & Ip, 2009; Wongsasaka, Chaijamrus, Kumkhong, & Boonanuntanasarn, 2015).

In the tricarboxylic acid cycle, α -ketoglutarate (AKG) can not only participate in a variety of amino to amino reactions, but also store and transport nitrogen to provide a carbon backbone for the synthesis of ornithine, glutamate, arginine and glutamine (Haghighat & McCandless, 2007; Harrison & Pierzynowski, 2008). Therefore, the addition of AKG can effectively improve the amino acid concentration and maintain the balance of amino acids in the body, thereby promoting the synthetic efficiency of proteins in the body and improving protein deposition (Junghans et al., 2006; Pierzynowski & Sjodin, 1998; Ren, Ai, & Mai, 2014). Additionally, in studies in animals, the administration of AKG has induced positive effects on strengthening immunity and antioxidant capacity (Le Boucher et al., 1997; Velvizhi, Nagalashmi, Essa, Dakshayani, & Subramanian, 2002). Moreover, in our previous study, dietary supplementation with 0.8% AKG solely in low-phosphorus diets can promote growth performance, P metabolism and the synthesis of collagen of Songpu mirror carp with an average initial body weight of 6.67 ± 0.08 g (Ai, Wang, Li, & Xu, 2019).

In this study, we aimed to investigate the possible protective effects of dietary AKG against various degrees of carbonate alkalinity stress on growth performance, antioxidant capacity and ammonia metabolism of Songpu mirror carp (*Cyprinus carpio* Songpu), which will further explore the mechanism of action of enzymes related to ammonia metabolism in high carbonate alkalinity environments and provide referential data for studying the application of AKG in saline aquaculture practice.

2 | MATERIALS AND METHODS

2.1 | Ethics statement

This study was performed according to the Guide for the Care and Use of Laboratory Animals in Heilongjiang River Fisheries Research Institute, Chinese Academy of Fishery Sciences, Harbin, China. All animal experiments were approved by the animal study ethical committee of Heilongjiang River Fisheries Research Institute, Chinese Academy of Fishery Sciences, and complied with Chinese laws, regulations and ethics.

2.2 | Experimental materials

Songpu mirror carp that were used in this experiment were purchased from Hulan Experimental Station of Heilongjiang Fisheries Research Institute (Harbin, China). AKG was provided by Sigma Company, and the purity of AKG was over 98%.

2.3 | Preparation of experimental diet

The formulation and proximate composition of the experimental diets are specified in Table 1. According to Formula feed for common carp (SC/T 1026–2002) (2002) and NRC (2011), fishmeal and soya bean meal were used as dietary protein sources and soya bean oil, fish oil and soya bean phospholipid were used as dietary lipid sources. According to the principle of equal nitrogen and equal energy, AKG was used to replace glucose in the basic feed, and two experimental diets containing 0% AKG and 1% AKG were prepared. All the ingredients were crushed and pulverized with an 80 mesh sieve. After weighing the ingredients according to the formula, all the ingredients were evenly mixed step by step and formed into 1-mm pellets using a pellet machine. The experimental diets were stored at -20°C until use.

2.4 | Experiment design

Two carbonate alkalinity gradients and one freshwater control group along with three parallel groups were designed for each treatment group. Sodium bicarbonate (analytical pure, product purity ≥ 99.5 ; Tianjin Zhiyuan Chemical Reagent Co., Ltd) was used to prepare the corresponding carbonate alkalinity, and 0.02 mmol/L HCl was used to calibrate it. Methyl orang-aniline blue (mixed) and phenolphthalein were used as indicators, and the measured carbonate alkalinities and pH are specified in Table 2.

2.5 | Feed administration

Songpu mirror carp were acclimatized to the experimental conditions and fed a commercial diet (Tongwei Co. Ltd., protein content: 36%, lipid content: 7%) three times daily for 2 weeks. After acclimation, the healthy carp with an average initial weight of 6.66 ± 0.08 g (mean $\pm$ SE) were randomly assigned to 18 aquaria with an initial stocking density of 30 fish per aquarium, and each diet was fed to three randomly assigned aquaria at 07:30, 12:30 and 16:30 daily. During 8-week feeding trial, feed was supplied at an amount of 5% body weight. The water quality parameters across the experiments were a temperature of $23.0\text{--}23.5^{\circ}\text{C}$, dissolved oxygen greater than 5.0 mg/L and ammonia nitrogen < 1.0 mg/L. During the trial, light was even. 20% of water was changed every day, and the fish tank was cleaned regularly to maintain good water quality. Aerated tap water was used to prepare the test water with different carbonate alkalinities.

2.6 | Sample collection

At the end of the trial, fish from each tank were harvested for 24 hr, counted and weighed (bulk and individually). Six fish were selected at random from each tank and were dissected to collect the blood, brain, gill filament, hepatopancreas, foregut, midgut and hindgut under moderate anaesthesia (MS-222 at 100 $\mu\text{g/kg}$). Blood samples

TABLE 1 Composition and nutrient levels of the basal diet (DM basis) %

Ingredients	Content
Fishmeal	8.0
Soya bean meal	56.0
Wheat middling	23.0
Fish oil	1.0
Soya bean oil	3.3
Phospholipid	1.0
Vitamin premix ^a	0.3
Mineral premix ^b	0.2
Choline chloride	0.5
$\text{Na}(\text{H}_2\text{PO}_4)_2$	2.0
Limestone	1.0
Methionine	0.3
Threonine	0.3
Cr_2O_3	0.1
CMCC	2.0
Glucose	1.50
Total	100.00
Nutrient levels	
Crude protein/CP ^c	38.38
Ether extract/EE ^c	6.23
Lys	2.20
Met	0.8
Thr	1.67
Total phosphorus/TP	0.85

^aThe vitamin premix provided the following per kg of the diet: VA 8,000 IU, VC 500 mg, VD_3 3,000 IU, VE 60 mg, VK_3 5 mg, VB_2 30 mg, VB_6 15 mg, VB_{12} 0.5 mg, choline chloride 5,000 mg, nicotinic acid 175 mg, D-biotin 2.5 mg, inositol 1,000 mg, folic acid 5 mg, pantothenic acid 50 mg.

^bThe mineral premix provided the following per kg of the diet: Zn 25 mg, Cu 3 mg, Fe 25 mg, Mn 15 mg, I 0.6 mg, Co 0.1 mg, Se 0.4 mg.

^cCP and EE were measured values, while the others were calculated values.

TABLE 2 Designed and measure carbonate alkalinity $N = 6$; means $\pm$ SD; mmol/L

Carbonate alkalinity and pH	Groups		
	Control	15 mmol/L	30 mmol/L
Carbonate alkalinity	0.00	15.00	30.00
Carbonate alkalinity	0.00	14.85	29.22
pH	6.95 ± 0.16	8.27 ± 0.16	8.31 ± 0.02

were first stored at 4°C for 1–2 hr and then centrifuged at 4°C and 3,500 r/min for 10 min to collect the supernatant (serum). The hepatopancreas, foregut, midgut and hindgut were stored at -20°C . The brain and gill filament were immediately placed into liquid nitrogen and stored at -80°C .

2.7 | Index determination method and analysis

2.7.1 | Determinations of growth parameters

On the last day of the trial, the fish were weighed ($W \pm 0.01$ g) and measured (body length, $L \pm 0.1$ cm). The collected data were used to calculate the following parameters:

$$\text{Weight growth rate (WGR, \%)} = 100\% \times \frac{(W_t - W_0)}{W_0}$$

$$\text{Feed conversion ratio (FCR)} = \frac{F}{(W_t - W_0)}$$

where W_t and W_0 are the final and initial body weights (g), respectively; t is the time of rearing (days); F is the total feed intake (g).

2.7.2 | Biochemical composition analysis

Six fish were selected at random and were used for whole-body composition analysis. Moisture was determined by oven-drying at 105°C to a constant weight. Crude protein was analysed by the Kjeldahl method after acid digestion. Crude lipids were determined by using the other extraction method with a Soxtec system. Crude ash was determined after combustion at 550°C to a constant weight.

2.7.3 | Measurement of the activity of antioxidant-related enzymes

The hepatopancreas, foregut, midgut and hindgut were homogenized in nine volumes of ice-cold physiological saline (0.86% w/v NaCl). The homogenates were centrifuged at 3,500 rpm for 15 min at 4°C (Thermo Fisher Scientific ST 16R centrifuge), and the resulting supernatants were aliquoted and stored at -20°C until subsequent analysis. The supernatant was used to determine superoxide dismutase (SOD, EC 1.15.1.1), catalase (CAT, EC 1.11.1.6) activity and glutathione (GSH) and soluble protein contents. All these indices were determined using commercial kits, which were produced by the Nanjing Jiancheng Bioengineering Institute, in Nanjing, China.

SOD and CAT activity was measured according to the SOD or GPx kit protocols (No. A001 or No. A007). GSH content was assayed according to a GSH kit protocol (No. A006).

2.7.4 | Determination of the serum ammonia content

The ammonia contents in serum were assayed according to the ammonia kit protocol (No. A086-1), which was produced by Nanjing Jiancheng Bioengineering Institute.

2.7.5 | Determination of the relative expressions of ammonia metabolism genes

The total RNA was isolated with RNAiso Plus according to the manufacturer's instructions. The purity and Quantitative OD260/280 of the isolated RNA were quantified by spectrophotometry at 260 and 280 nm, respectively, whereas the quality of the RNA was verified by agarose gel electrophoresis. The first-strand cDNA was synthesized with the TaKaRa PrimeScript™ RT Reagent Kit with gDNA Eraser (Perfect Real Time) (Code No: RR047A) (Dalian Takara Company).

Real-time PCR was conducted with the cDNA template and the TaKaRa SYBR® Premix Ex Taq™ (Tli RNaseH Plus) (Code No: RR420A) (Dalian Takara Company) following the manufacturer's protocol. The primers are listed in Table 3. β -Actin was selected as the internal standard because of its stable expression in different tissues over time. The $2^{-\Delta\Delta C_t}$ method was used to determine the relative expression levels of each gene.

2.8 | Data processing and analysis

The relationship between AKG and ammonia levels was determined by two-way ANOVA, using the levels of carbonate alkalinity and dietary AKG as two factors. If a significant effect was detected, differences between group means were compared by Duncan's multiple range tests. The level of significance was set at $p < .05$, and the results were expressed as means $\pm$ standard deviation (SD). All analyses were performed using the SPSS 19.0 for Windows.

TABLE 3 Fluorescence quantitative primer sequences

Positive primer	Primer sequence 5'-3'	Reaction primer	Primer sequence 5'-3'
AQP3A	GGACTGGCTGAATGCTTGG	AQP3A	CTGGCCACAAACCAAGATTC
Rhag	AGCAGTGTGGGCATCAACCT	Rhag	CTCCAGCAGGGTCATGATGAG
Rhcg1	AGTCTGACACTCGCTGGATCG	Rhcg1	AGCAGAGCCCACTGAATTCC
Rhcg2	CAATGGGCTTTGCTGATGC	Rhcg2	AGCGCCATAAGCAATGAGAC
NKA	ATACTGATGCTGTGTGGCTTG	NKA	CAGGAACGATGAAGCGATG
GS	CCTGTCTGTATGCTGGGGTA	GS	CAAAGTCCTCACAAACCCTG
Actin-F	GATTGGCTGTGAGATGATGCT	Actin-R	CGTTGTAGAAGGTGTGATGCC

3 | RESULTS

3.1 | Growth performance and feed utilization

The growth parameters and feed utilization are presented in Table 4. One per cent AKG can significantly improve the weight growth rate (WGR) ($p < .05$) and reduce the feed conversion ratio (FCR) ($p > .05$) at a carbonate alkalinity of 0 mmol/L. In addition, according to two-way ANOVA, the WGRs were significantly influenced by the carbonate alkalinity levels and AKG ($p < .05$), whereas the FCR was significantly influenced by AKG ($p < .05$) only.

3.2 | Body composition

As seen in Table 5, moisture, crude protein and crude lipid were not significantly affected by either the carbonate alkalinity or the dietary AKG levels ($p > .05$). Crude ash was significantly higher in fish fed diets containing 1% AKG compared with that of fish fed with diets containing 0% AKG at the carbonate alkalinities of 15 mmol/L and 30 mmol/L ($p < .05$). A significant interaction between the carbonate alkalinity and dietary AKG levels was found for ash ($p < .05$).

3.3 | Antioxidant capacity

Table 6 presents the changes in the activities of superoxide dismutase (SOD) of Songpu mirror carp. According to two-way ANOVA, a significant effect was observed in which increasing the carbonate alkalinity significantly decreased the SOD activity in the foregut and hepatopancreas ($p < .05$). Additionally, at a carbonate alkalinity of 0 mmol/L, 1% AKG significantly increased the SOD activity ($p < .05$). Significant effects of the interaction between the carbonate alkalinity and dietary

AKG levels on the SOD activity in the hepatopancreas were observed ($p < .05$).

Table 7 presents the changes in the activities of catalase (CAT) of Songpu mirror carp. According to two-way ANOVA, increasing the carbonate alkalinity significantly decreased the CAT activity in the foregut and hepatopancreas ($p < .05$), and AKG significantly reduced CAT activity in the hepatopancreas. Additionally, at a carbonate alkalinity of 0 mmol/L, 1% AKG significantly improved the CAT activity in the foregut ($p < .05$), whereas at a carbonate alkalinity of 15 mmol/L, 1% AKG significantly decreased the CAT activity in the hepatopancreas ($p < .05$).

Table 8 presents the changes in the contents of glutathione (GSH) of Songpu mirror carp. According to two-way ANOVA, the content of GSH in the foregut and hepatopancreas significantly decreased with increasing alkalinity levels ($p < .05$). In addition, diets with AKG reduced the content of GSH in the foregut, midgut, hindgut and hepatopancreas, although there were no significant differences ($p > .05$).

3.4 | Ammonia metabolism

3.4.1 | Blood ammonia

As seen in Figure 1, fish fed at carbonate alkalinities of 15 mmol/L and 30 mmol/L exhibited a significantly higher blood ammonia content compared with fish fed at a carbonate alkalinity of 0 mmol/L ($p < .05$). No significant difference in the blood ammonia content was noted among fish fed different AKG levels ($p > .05$). It was notable that at carbonate alkalinities of 15 mmol/L and 30 mmol/L, fish fed diets containing 1% AKG exhibited a significantly higher blood ammonia content compared with fish fed diets containing no AKG, although no significant difference was observed ($p > .05$).

TABLE 4 Growth performance and feed utilization of Songpu mirror carp under different treatments

Carbonate alkalinity levels(mmol/L)	AKG levels(%)	Initial weight/g	Weight gain rate (WGR)/%	Feed conversion ratio(FCR)
0	0	6.63 ± 0.03	773 ± 23b	1.37 ± 0.03ab
	1	6.62 ± 0.12	859 ± 36a	1.32 ± 0.06b
15	0	6.66 ± 0.15	506 ± 19cd	1.47 ± 0.0ab
	1	6.68 ± 0.01	541 ± 6c	1.63 ± 0.14a
30	0	6.66 ± 0.01	485 ± 11d	1.57 ± 0.06ab
	1	6.68 ± 0.11	512 ± 5cd	1.60 ± 0.10a
p-value of two-way ANOVA				
AKG			.001	.481
Carbonate alkalinity			.000	.025
Carbonate alkalinity* AKG			.132	.462

Note: All the values are expressed as the mean values ± SD, and different letters in the same column denote a significant difference ($p < .05$).

*** means the interaction of Carbonate alkalinity and AKG.

TABLE 5 Body composition of Songpu mirror carp under different treatments

Carbonate alkalinity levels(mmol/L)	AKG levels(%)	Moisture/%	Crude protein/(% w.w.)	Crude lipid/(% w.w.)	Ash/(% w.w.)
0	0	73.9 ± 0.5	16.5 ± 0.2	7.59 ± 0.47	2.69 ± 0.90b
	1	74.9 ± 0.2	15.8 ± 0.3	6.97 ± 0.13	2.72 ± 0.17b
15	0	75.2 ± 1.0	16.3 ± 0.7	6.20 ± 0.64	2.72 ± 0.16b
	1	74.5 ± 0.7	15.9 ± 0.6	6.63 ± 0.45	3.09 ± 0.14a
30	0	74.1 ± 0.4	16.5 ± 0.3	6.99 ± 0.17	2.80 ± 0.11b
	1	74.5 ± 1.1	15.7 ± 0.6	6.65 ± 0.57	3.37 ± 0.27a
<i>p</i> -value of two-way ANOVA					
AKG		.667	.103	.496	.000
Carbonate alkalinity		.339	.804	.104	.645
Carbonate alkalinity* AKG		.174	.642	.329	.006

Note: All the values are expressed as the mean values ± SD, and different letters in the same column denote a significant difference ($p < .05$).

*** means the interaction of Carbonate alkalinity and AKG.

TABLE 6 Activities of SOD of Songpu mirror carp under different treatments (U/mg prot)

Carbonate alkalinity levels(mmol/L)	AKG levels(%)	Foregut	Midgut	Hindgut	Hepatopancreas
0	0	75.8 ± 10.6a	54.6 ± 8.1	72.2 ± 4.1	25.5 ± 1.4b
	1	66.1 ± 4.1ab	56.2 ± 10.0	70.3 ± 6.5	30.3 ± 2.3a
15	0	50.16 ± 1.86c	70.4 ± 7.1	61.7 ± 2.7	31.7 ± 0.7a
	1	61.5 ± 1.5bc	53.9 ± 2.8	74.1 ± 5.6	25.1 ± 1.1b
30	0	56.4 ± 1.9bc	57.5 ± 2.5	73.7 ± 4.7	24.9 ± 1.9b
	1	55.7 ± 4.5bc	51.1 ± 4.2	70.4 ± 7.2	23.3 ± 0.7b
<i>p</i> -value of two-way ANOVA					
AKG		.928	.198	.579	.345
Carbonate alkalinity		.006	.416	.536	.025
Carbonate alkalinity* AKG		.076	.403	.480	.008

Note: *** means the interaction of Carbonate alkalinity and AKG.

All the values are expressed as the mean values ± SD, and different letters in the same column denote a significant difference ($p < .05$).

3.4.2 | Relative expression of glutamine synthase (GS) in the brain

As seen in Figure 2, the relative expression of glutamine synthase (GS) in the brain of mirror carp was not affected by the carbonate alkalinity levels ($p > .05$). Moreover, the relative expression of GS in the brain of Songpu mirror carp was significantly increased by the dietary AKG levels, with the highest observed in fish fed a diet containing 1% AKG at a carbonate alkalinity of 30 mmol/L ($p < .05$).

3.4.3 | Relative expressions of ammonia metabolism genes in the gills

As seen in Figure 3, increasing the carbonate alkalinity can significantly improve the relative expression of AQP3, Rhag, Rhcg2 and Na^+/K^+ -ATPase (NKA) in the gills ($p < .05$). Additionally, diets with 1% AKG

can decrease the relative expression of AQP3 and Na^+/K^+ -ATPase (NKA) in gills, although there was no significant difference ($p > .05$).

4 | DISCUSSION

4.1 | Growth performance, feed utilization and body composition

Normal growth and reproduction of aquatic animals require a certain carbonate alkalinity to maintain the buffer capacity and to improve the productive forces of the water environment, but at the same time, it should be noted that growth will be inhibited beyond the appropriate range (Danalat & Selcuk, 1992; Eva & Stephan, 1992). Jiang obtained results from the study on the *Chalcalburnus chalcoides* in which the optimum carbonate alkalinity range was 7.36–21.46 mmol/L (Jiang, Lin, Wang, & Liu, 2008). For *Allogynogomphus*

TABLE 7 Activities of CAT of Songpu mirror carp under different treatments (U/mg prot)

Carbonate alkalinity levels(mmol/L)	AKG levels(%)	Foregut	Midgut	Hindgut	Hepatopancreas
0	0	45.6 ± 3.8b	74.5 ± 4.0a	53.3 ± 8.2ab	16.1 ± 1.2a
	1	61.7 ± 3.2a	65.4 ± 13.4ab	55.4 ± 10.9a	13.2 ± 0.2ab
15	0	48.3 ± 3.4ab	53.9 ± 4.3bc	26.9 ± 1.2c	16.2 ± 1.7a
	1	49.8 ± 4.4ab	43.5 ± 0.7c	37.1 ± 5.6bc	11.9 ± 1.2bc
30	0	37.9 ± 5.5b	50.0 ± 4.1bc	20.8 ± 2.5c	11.8 ± 0.5bc
	1	39.8 ± 5.3b	42.8 ± 6.5c	23.1 ± 1.6c	9.3 ± 0.2c
<i>p</i> -value of two-way ANOVA					
AKG		.091	.090	.300	.002
Carbonate alkalinity		.015	.006	.000	.003
Carbonate alkalinity* AKG		.203	.957	.710	.626

Note: All the values are expressed as the mean values ± SD, and different letters in the same column denote a significant difference ($p < .05$).

*** means the interaction of Carbonate alkalinity and AKG.

TABLE 8 Contents of GSH of Songpu mirror carp under different treatments (mgGSH/g prot)

Carbonate alkalinity levels(mmol/L)	AKG levels(%)	Foregut	Midgut	Hindgut	Hepatopancreas
0	0	25.9 ± 3.7a	17.5 ± 7.5	17.6 ± 4.7	20.6 ± 2.3a
	1	20.0 ± 3.4ab	20.5 ± 5.6	12.7 ± 2.6	17.2 ± 3.8ab
15	0	10.1 ± 0.4c	15.4 ± 6.0	11.9 ± 1.8	13.1 ± 2.4b
	1	15.5 ± 3.2bc	16.7 ± 0.8	17.1 ± 5.3	10.0 ± 2.0b
30	0	13.2 ± 2.8bc	19.0 ± 3.8	16.7 ± 0.7	9.9 ± 1.4b
	1	11.9 ± 1.6bc	16.4 ± 2.6	15.2 ± 4.5	9.8 ± 1.2b
<i>p</i> -value of two-way ANOVA					
AKG		.768	.884	.933	.245
Carbonate alkalinity		.004	.830	.892	.007
Carbonate alkalinity* AKG		.119	.820	.383	.687

Note: All the values are expressed as the mean values ± SD, and different letters in the same column denote a significant difference ($p < .05$).

*** means the interaction of Carbonate alkalinity and AKG.

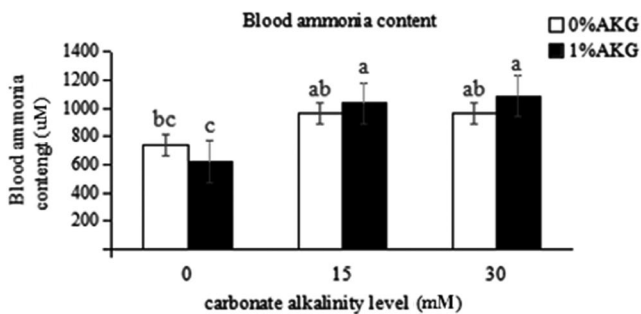


FIGURE 1 Blood ammonia contents of Songpu mirror carp under two different treatments. Note: All values are expressed as the mean values ± SD, and different letters in the figure denote a significant difference ($p < .05$)

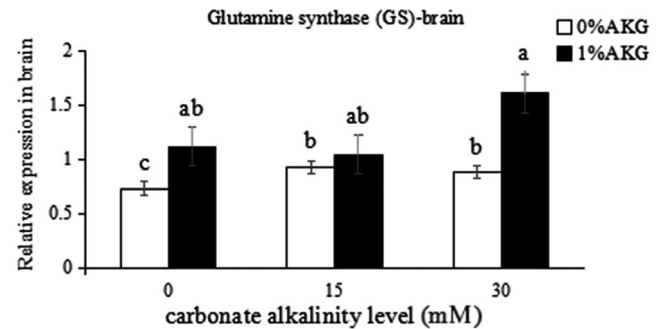


FIGURE 2 Relative expression of glutamine synthase (GS) in the brain of Songpu mirror carp under different treatments. Note: All values are expressed as the mean values ± SD, and different letters in the figure denote a significant difference ($p < .05$)

crucian, a significant reduction in the feed ration was found when the carbonate alkalinity reached 10 mmol/L (Lv, Huang, & Wang, 2007). Our study demonstrated that, irrespective of the dietary AKG level, the WGR and FCR of fish fed at carbonate alkalinities of 15 mmol/L

and 30 mmol/L were lower than those of fish fed at a carbonate alkalinity of 0 mmol/L, which indicates that the optimum carbonate alkalinity range of Songpu mirror carp is below 15 mmol/L. The influence mechanism of carbonate alkalinity on the growth of fish may

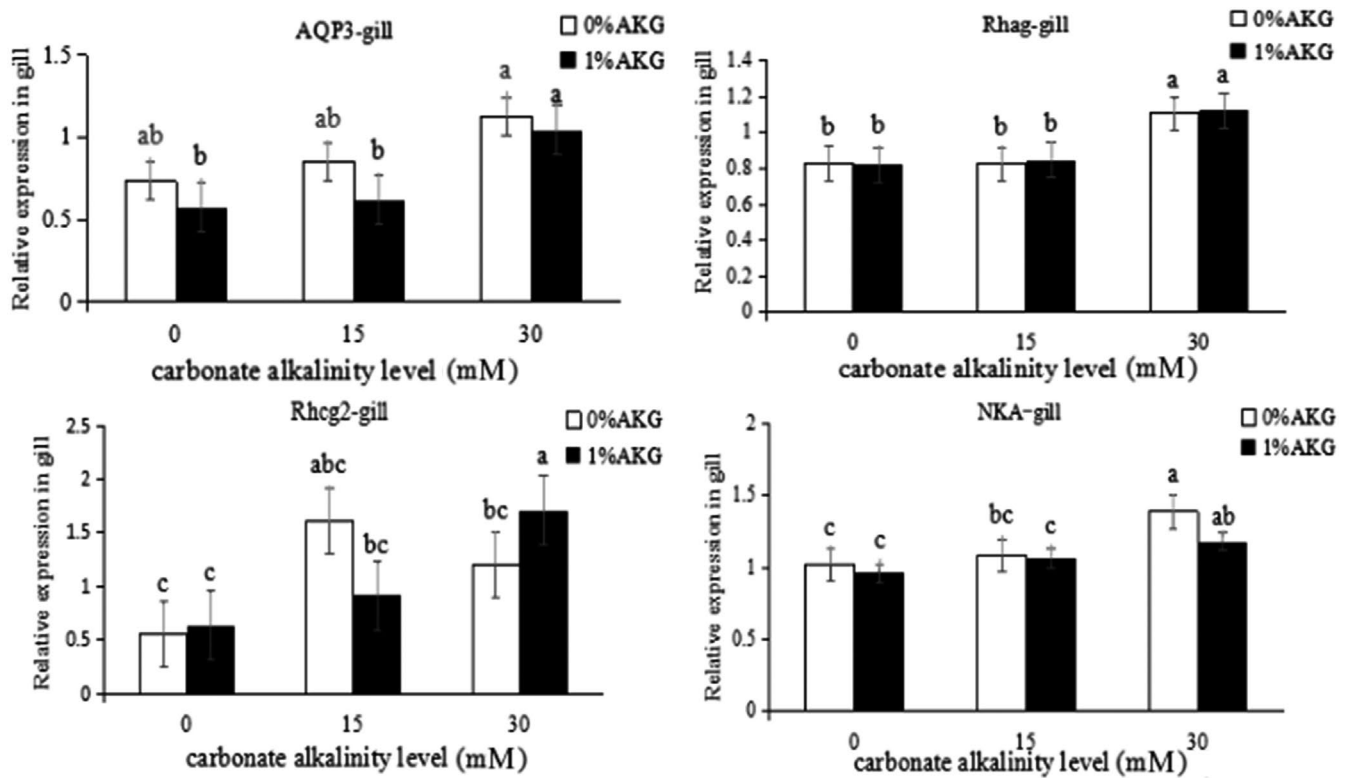


FIGURE 3 Relative expression of AQP3, Rhag, Rhcg2 and NKA in the gills of Songpu mirror 29 carp under different treatments. Note: All values are expressed as the mean values $\pm$ SD, and different letters in the figure denote a significant difference ($p < .05$)

be embodied in many aspects, among which a decrease in digestion ability and a change in energy distribution are two important effects. On the one hand, increasing the carbonate alkalinity can increase the pH in water, thus influencing the pH in the gut, so that the activities of digestive enzymes will decrease when the pH in the gut is beyond the optimum pH of the digestive enzymes, which inhibits the food ration of fish (Fang & Chiou, 1989; Lv et al., 2007; Wei et al., 2013). On the other hand, increasing the carbonate alkalinity can also change the energy distribution, so that fish have to consume more energy to regulate the osmotic pressure, which inhibits feed conversion (Hiroi & McCoyrmick, 2012). Therefore, further studies concerning the optimum carbonate alkalinity range and the adaptive mechanism for Songpu mirror carp ought to be designed.

Previous studies on the effects of the dietary AKG levels mainly focused on livestock and poultry animals, including weaned pig (Liu et al., 2009), rat (Dobrowolsk et al., 2008) and weaned rabbit (Chamorro et al., 2010), while relatively few dealt with aquatic animals. Therefore, in this study, we aimed to estimate the effect of AKG on fish. Our results revealed that 1% AKG can significantly improve the WGR and reduce the FCR of Songpu mirror carp compared with no AKG, but there was no difference between the two levels of groups. However, different results were observed in the previous study (Wang et al., 2016), which reported that a 1% addition of AKG can significantly enhance the WGR and reduce the FCR of hybrid sturgeon (*Acipenser schrenckii* $\times$ *A. baeri*), which might be attributed to the differences among the fish species, breeding environments, growth phases, dietary nutrient levels, and so on.

However, despite some differences among the results of different experiments, there is one thing that cannot be denied—AKG can promote the growth of the fish body. Two explanations can be used to illustrate this mechanism of action of AKG. First, the addition of AKG can promote the development of the intestinal tract, enhance the digestion and absorption rate of nutrients, and further boost the growth performance. Previous study has found that the addition of AKG can reduce mucosal damage and improve the absorption function of the small intestine, thereby improving the growth performance of piglets (Hou et al., 2010). Second, as a precursor of glutamine, AKG can transport and store nitrogen. After entering the body, AKG can effectively maintain the total nitrogen balance and protein synthesis and further promote the growth (Kristensen, Jungvid, Fernandez, & Pierzynowski, 2002; Xiao et al., 2016). Hence, future studies should aim to collect more data on the corresponding relationship between the dietary AKG level and the carbonate alkalinity level in the water environment.

Our results showed that the body composition was not significantly affected by the carbonate alkalinity of water, which suggested that carbonate alkalinity has no significant effect on nutrient metabolism and deposition, but the empirical analyses need to be strengthened. Related studies have demonstrated that animals fed a diet containing a certain concentration of AKG can show a preferable nutrient synthesis capacity (Yao et al., 2012). However, our results showed that the crude protein and lipid contents of fish were relatively stable with increased dietary AKG levels, which is consistent with the finding (Wei et al., 2013), which indicated that there was

no significant difference in the body composition of Songpu mirror carp among the groups of dietary AKG levels, therefore suggesting that the nutrients were not used for nutrient deposition. Different results were observed in the study of Songpu mirror carp (Wei et al., 2015), which may be due to the differences in the growth phase and breeding environment.

4.2 | Antioxidant capacity

Antioxidant capacity is an important indicator to reflect the health status of organisms. For fish, hepatopancreas and intestine are two vital organs to address the oxidative injury (Rust, 2002). However, it has been known that fish intestine can be subdivided into three segments (Rombout, Abelli, Picchietti, Scapigliati, & Kiron, 2011). In common carp (*C. carpio* L.), the hindgut has much higher number of granulocytes and large Ig-positive macrophages than those in the foregut and midgut (Rombout, Anja, Thiele T, & Villena, 1993), indicating that the immunocompetence in each intestinal segment is somewhat different. Therefore, in addition to monitoring the antioxidant capacity of major antioxidant organ—hepatopancreas, this study also monitors the antioxidant capacity of the three segments of intestine. The antioxidant system includes the enzyme system (SOD, CAT, POD, et al.) and the coenzyme system (GSH, VC, VE, et al.), which will be affected by the breeding environment and the feed composition (Bagnyukova, Chahrak, & Lushchak, 2006).

Previous studies have frequently demonstrated that environmental stress can cause the accumulation of reactive oxygen in fish, which leads to oxidative damage, thereby posing a threat to the health of fish (Schreck, Contreras-Sanchez, & Fitzpatrick, 2001; Volodymyr & Tetyana, 2007). Our results revealed that after the 8-week feeding trial, significant decreases in the SOD activity in the hepatopancreas and foregut; the CAT activity in the hepatopancreas, foregut, midgut and hindgut; and the GSH content in the hepatopancreas and foregut of fish fed at alkalinities of 15 mmol/L and 30 mmol/L were observed compared with fish fed at a carbonate alkalinity of 0 mmol/L, which were consistent with a previous finding that dramatic changes in the salinity and chloride ion concentration in water caused a remarkable decrease in the antioxidant ability of *Dicentrarchus labrax* (Guerriero, Di, & Ciarcia, 2002). In *Gymnocypris przewalskii* exposed to high concentrations of carbonate alkalinity for 9 days, the activity of the related antioxidant enzyme first showed an elevation under stress and then returned to the initial level (Wang et al., 2013). The above difference might be mainly attributed to the different experimental cycle. Moreover, the difference in the above research results may suggest that the chronic stress of high-concentration carbonate alkalinity will produce higher oxidative stress, thus affecting the metabolic response in organisms.

Glutamine is an important precursor of glutathione synthesis, and AKG is a precursor of glutamine, suggesting that the appropriate AKG level can enhance the antioxidant ability of the body (Dong, Wei, Yu, Wang, & Xu, 2014). A significant improvement in

the content of GSH in the hepatopancreas of hybrid sturgeon (*A. schrenckii* ♀ × *A. baeri* ♂) was observed in fish offered a diet containing AKG (Chen, Wang, & Xu, 2015). A different phenomenon was observed in our study, as the GSH contents of the hepatopancreas, foregut, midgut and hindgut were not significantly affected by increased dietary AKG levels, which indicated that AKG may not affect the antioxidant capacity of Songpu mirror carp by regulating non-enzymatic systems. Similarly, AKG addition in our laboratory has been shown to benefit Songpu mirror Carp in terms of antioxidant capacity (Song, Wang, & Xu, 2016). Meanwhile, our results showed that the higher the SOD activities of hepatopancreas and foregut and CAT activity of the hepatopancreas were observed in fish fed a diet containing 1% AKG. In addition, it must be noted that at a carbonate alkalinity of 15 mmol/L, fish fed a diet containing 1% AKG showed higher SOD and CAT activities and higher GSH contents in the foregut, midgut and hindgut, although there was no significant difference among all groups. This observation is in accordance with previous findings (Wang et al., 2017), which reported that dietary supplementation with 1.0% AKG increased the SOD and CAT activities in the serum and intestine of hybrid sturgeon (*A. schrenckii* ♀ × *A. baeri* ♂). These findings may suggest that AKG can mitigate the oxidative damage caused by carbonate alkalinity stress to a certain extent. Nonetheless, appropriate AKG levels corresponding to different carbonate alkalinities need more in-depth study.

4.3 | Ammonia metabolization

For fish, ammonia is a normal metabolite in vivo that plays an irreplaceable role in the maintenance of the acid-base balance. In a high-concentration carbonate alkalinity environment, the passive diffusion of NH_3 will be inhibited, thus leading to NH_3 accumulation and thereby producing ammonia poisoning in vivo (Randall & Tsui, 2002). *Leuciscus waleckii*, living in Dali Lake, which has high carbonate alkalinity, had higher contents of blood ammonia than that of those living in Gang Geng Lake, which has low carbonate alkalinity (An, Qing, Luo, Wu, & Wang, 2014). A similar phenomenon was observed in the present study, as the blood ammonia content was significantly elevated during the 8-week carbonate alkalinity administration. To address ammonia accumulation and to alleviate the toxic effects of ammonia accumulation in vivo, fish may possess two ammonia metabolism pathways: one is the urea metabolism pathway and the other is the glutamine metabolism pathway, which is the main pathway of ammonia metabolism (Tu et al., 2019). According to the literature on rainbow trout (Sanderson, Wright, Robinson, Ballantyne, & Bernier, 2010), *Clarias gariepinus* (Wee et al., 2007), glutamine plays a crucial role in the degradation of ammonia nitrogen toxicity in the brain. Glutamine is synthesized and released from brain tissue into the bloodstream and hepatopancreas as a non-toxic ammonia metabolite. Moreover, in response to the high ammonia concentration in vivo, the ion channels (NKA, et al.), protein channels (AQP family, et al.) and specialized protein carriers for ammonia transfer (Rh proteins, et al.) will participate in the ammonia transport process (Agre

& Kozono, 2003; Chew & Ip, 2014; Liu, Nagase, Huang, Calamita, & Agre, 2006; Tay et al., 2006). Our result revealed that, irrespective of the dietary AKG levels, the gene expression of AQP3, Rhag, Rhcg2 and NKA in the gills and GS in the brain was significantly upregulated with increasing carbonate alkalinity levels, which were positively correlated with the blood ammonia contents, meaning that Songpu mirror carp might simultaneously trigger the glutamine metabolism pathway and enhance the gene expression of proteins related to the ammonia transfer, thereby adjusting the blood ammonia concentration in the chronic stress of high-concentration carbonate alkalinity. Similar findings were also found in *Danio rerio* (Braun, Steele, & Perry, 2009), *C. carpio* (Zhao, Xu, & Sun, 2013), *L. waleckii* (Chang et al., 2014) and *Oreochromis niloticus* (Tu et al., 2019).

In terms of dietary AKG levels, our result manifested that in the freshwater group, fish in the 1% AKG group showed lower blood contents compared with those that were fed the no AKG group, which was consistent with a previous finding that 1.5% AKG increased the blood ammonia content of Songpu mirror carp (Wei et al., 2015). Returning to the carbonate alkalinity groups, the change showed a remarkable opposite trend compared to that of the freshwater group, which suggested that 1% AKG cannot effectually alleviate the NH₃ accumulation caused by chronic carbonate alkalinity stress, which may be due to the circumstance of increased carbonate alkalinity in the water environment inhibiting the utilization of AKG for fish in our experiment. From the perspective of ammonia metabolism and transfer, our result showed that with the increasing AKG levels, there was significant upregulation of the gene expression involved in GS in the brain, whereas there was significant downregulation of the gene expression involved in AQP3, Rhag, Rhcg2 and NKA in the gills. Similar findings were also found in the study on Songpu mirror carp (Zhao, Song, & Xu, 2017) and hybrid sturgeon (*A. schrenckii* ♀ × *A. baeri* ♂) (Wang et al., 2017). The reason might be that AKG can be converted into glutamine by glutamine synthase, which not only provides abundant glutamine for protein synthesis but also reduces the damage of ammonia nitrogen to the body (Zheng, Ruan, Xu, & Yang, 2008). Consequently, the above result may testify that under the chronic stress of high-concentration carbonate alkalinity, AKG can boost the glutamine metabolism pathway to achieve the response strategy of a reduction in ammonia and reveal a weakness in promoting ammonia transfer.

5 | CONCLUSION

Chronic carbonate alkalinity stress can significantly inhibit the growth and weaken the antioxidant capacity and ammonia metabolism of Songpu mirror carp, whereas addition of AKG can enhance the regulation of the expression of ammonia metabolism and improve the antioxidant ability of Songpu mirror carp in a high-concentration carbonate alkalinity environment. The results of this study provide a theoretical basis for the application of AKG as a feed additive for aquatic animals living in an environment of chronic stress caused by high-concentration carbonate alkalinity.

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CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest.

AUTHORS' CONTRIBUTIONS

All authors contributed to experimental design, data acquisition and analysis, as well as manuscript preparation and editing.

ORCID

Ze Fan  <https://orcid.org/0000-0002-1365-0016>

Qiyu Xu  <https://orcid.org/0000-0001-8380-2022>

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